Automatic Reference Counting(ARC) and memory mangement

```
1.class Person {
    let name: String
    var apartment: Apartment?

    init(name: String) {
        self.name = name
    }

    deinit {
        print("Person \(name) deallocated")
    }
}

class Apartment {
    let unit: String
    weak var tenant: Person?

    init(unit: String) {
        self.unit = unit
    }

    deinit {
        print("Apartment \(unit) deallocated")
    }
}

var person: Person? = Person(name: "Raja")
var apartment: Apartment? = Apartment(unit: "A101")

person?.apartment = apartment
apartment?.tenant = person
```

person = nil

apartment = nil

Output

```
Person Raja deallocated
Apartment A101 deallocated
```

2. class Downloader {

```
    let name: String
    var onFinish: (() -> Void)?

    init(name: String) {
        self.name = name
    }

    func start() {
        onFinish = { [weak self] in
            guard let self = self else { return }
            print("Finished \(self.name)")
        }

        onFinish?()
    }

    deinit {
        print("Downloader deallocated")
    }
}
```

var downloader: Downloader? = Downloader(name: "File Download")

downloader?.start()

downloader = nil

Output

```
Finished File Download
Downloader deallocated
```

3. class Box {

    deinit {

        print("Box gone")

    }

}

var a: Box? = Box()

var b = a

var c = a

a = nil

b = nil

print("before c")

c = nil

print("end")

Output

```
before c
Box gone
end
```